

Calibration of an Eye-to-Hand System Using a Laser Pointer on Hand and Planar Constraints

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Abstract—This work proposes a technique for calibration of an eye-to-hand system. The target of the hand-eye calibration is to estimate the geometric transformation between the hand and the eye. This calibration method further considers camera intrinsic parameters and geometric relations of a working plane in space at the same time. A laser pointer casually mounted on the hand is utilized. By manipulating the robot and projecting the laser beam on a plane of unknown orientations, a batch of related image positions of light-spots are extracted from images of the camera. Since the laser is rigidly mounted and the plane is fixed at each orientation, the geometric parameters and measurement data must obey a certain nonlinear constraints and the solutions of parameters can be estimated accordingly. A close-form solution is developed by decoupling the nonlinear equations into linear forms to compute all of the initial values. As a result, the calibration method does not need any manual initial guess of the unknown parameters. To achieve a higher accuracy, a nonlinear optimization method is implemented to refine the estimation. The advantage of using laser pointer is that this technique can be used for the case when the eye does not see the hand. Experimental results of simulations and real data are presented to show the validity and the simple requirements of the proposed algorithm.

I. INTRODUCTION

Hand-eye coordination system can provide reliability and flexibility for complex tasks. Eye indicates the camera while hand indicates the robot manipulator in such system. Hand-eye coordination system can be classified into two types of different camera configurations, namely eye-in-hand and eye-to-hand system. A camera rigidly mounted on the end-effector (eye-in-hand) is more maneuverable to observe the target in detail. On the other hand, the static camera (eye-to-hand) has global view and can detect the changes of the scene more easily. Most works consider one configuration for visual servo. Flandin *et al.* [1] further introduce a cooperation of eye-in-hand and eye-to-hand to take both advantages. For position based visual servo, calibrations of the camera and the transformation between the camera and the robot are necessary. Since the pose of control points are

estimated from their projections on the image from the camera, positioning error of a target may be serious due to inaccurate camera parameters. To determine the geometric relationship is referred as hand-eye calibration. Positioning of the end-effector may be biased due to errors in this relationship. It is possible to correct such errors by directly observing the end-effector. However, in many cases, control points on the end-effector and target cannot be observed together. So there is no guarantee on the positioning accuracy of the visual servo system [2]. Therefore, it is still important to find new ways to improve the accuracy of hand-eye calibration.

Hand-eye calibration of the eye-in-hand configuration is discussed from mid eighties until recent years and some solutions are provided for accurately estimating parameters [3]-[9]. Most of them assume the camera is well calibrated and aim at calibration of geometric relations by utilizing 3D object, 2D pattern, or just a point. Strobl and Hirzinger further take the camera model into the relationship and their method can relax the need of accurate grid pattern dimensions [9]. Calibration of the eye-to-hand configuration was rarely discussed since the relationship is in the same form of eye-in-hand [5]. Other researches focus on robot-world coordination [10] through specially made devices.

In some cases, the eye does not see the hand for coordination. For example, in applications of product sorting on a moving conveyor, the camera often placed at a distance away from the arm to compensate the delay of image processing and to avoid interference of target tracking on a fast moving conveyor belt. In catching ball system, e.g. [11], cameras should focus on the region of initial ball flying trajectory and they might be able to see the arm. Lei *et al.* [10] introduce a calibration method for robot-world with a triple laser device and their method may be a solution under this eye-to-hand situation. But they need a special designed laser device with a pre-calibrate camera and eye-world relation.

This work develops a method for eye-to-hand system which can simultaneously accomplish camera calibration, hand-eye calibration, and coordination of the working plane related to the robot base frame. This method relaxes the requirement of precise calibration pattern under the accuracy of the manipulator. It can work for the situation of that the hand is not in the view field of the eye. In other words, it provides the ability for a longer working range of hand-eye coordination. Only two extra devices are required, i.e., a laser emitter and a planar object. In this work, the initial guess problem for nonlinear optimization is further solved and this

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allows the proposed method to work automatically. The rest of this paper is structured as follows. Section II describes the parameters for calibration and their notations. Section III presents in detail how to solve the calibration problem with a plane and a laser. Both close-form solution and nonlinear optimization based on the constraint are also proposed in this section. Section IV provides experimental results by simulation and real data. Finally, Section V concludes the paper.

II. PARAMETERS FOR CALIBRATION AND NOTATIONS

This section introduces parameters of the eye-to-hand system for calibration including the intrinsic and extrinsic parameters of the camera and the geometric relations of the working plane related to the robot base coordinate system.

1) *Intrinsic parameters of camera*: The camera model is as pin-hole one with considering lens distortion. A 2D position in a image is denoted by $\mathbf{m} = [u, v]^T$ and its homogeneous coordinate is defined by $\tilde{\mathbf{m}} = [u, v, 1]^T$. A 3D position of Cartesian coordinate system in the camera frame is denoted by $\mathbf{x}_c = [x_c, y_c, z_c]^T$.

$$\tilde{\mathbf{m}} = \mathbf{K} \cdot \begin{bmatrix} x_d \\ y_d \\ 1 \end{bmatrix} \quad \text{with } \mathbf{K} = \begin{bmatrix} f_u & \alpha_c \cdot f_u & u_0 \\ 0 & f_v & v_0 \\ 0 & 0 & 1 \end{bmatrix} \quad (1)$$

where $\mathbf{K}$ is the intrinsic matrix including five intrinsic parameters, and

$$\begin{bmatrix} x_d \\ y_d \end{bmatrix} = \left(1 + \kappa_1 \|\mathbf{x}_n\|^2 + \kappa_2 \|\mathbf{x}_n\|^4\right) \mathbf{x}_n + \begin{bmatrix} 2\rho_1 x_n y_n + \rho_2 (\|\mathbf{x}_n\|^2 + 2x_n^2) \\ \rho_1 (\|\mathbf{x}_n\|^2 + 2y_n^2) + 2\rho_2 x_n y_n \end{bmatrix} \quad (2)$$

in that $\mathbf{x}_n \triangleq [x_c/z_c \quad y_c/z_c]^T$, κ_1 and κ_2 are two radial distortion parameters, and ρ_1 and ρ_2 are two tangential distortion parameters. There are total nine intrinsic parameters of this camera model. For the pinhole model without considering distortion, i.e., all distortion parameters are zeros, we have

$$\mathbf{x}_d^T = \mathbf{x}_n^T = [x_c/z_c \quad y_c/z_c]^T. \quad (3)$$

2) *Eye-to-Hand transformation*: Transformation from camera frame to robot base frame is determined by a rotation matrix ${}^B_c\mathbf{R}$ and a translation vector ${}^B_c\mathbf{T}$. A point ${}^c\mathbf{p}$ of 3D Cartesian coordinate system in camera frame transforms into robot base coordinate system via,

$${}^B\mathbf{p} = {}^B_c\mathbf{R} \cdot {}^c\mathbf{p} + {}^B_c\mathbf{T}, \quad (4)$$

The rotation matrix is derived by the direction cosine matrix, $\mathbf{R}(\theta, \phi, \varphi)$. The rotation sequence is Z-Y-X, where θ is z-axis rotation, ϕ is y-axis rotation, and φ is x-axis rotation. In following context, the robot base coordinate system is identical to the world coordinate of the whole system. The transformation from end-effector frame to robot base frame is determined by a rotation matrix ${}^B_e\mathbf{R}$ and a translation

vector ${}^B_e\mathbf{T}$.

A homogeneous matrix from robot base coordinate system to camera coordinate system is defined as

$$\mathbf{G} = \begin{bmatrix} g_{11} & g_{12} & g_{13} \\ g_{21} & g_{22} & g_{23} \\ g_{31} & g_{32} & g_{33} \end{bmatrix} \triangleq \mathbf{K} \cdot {}^c_B\mathbf{R} \quad (5)$$

3) *Working plane*: The planar object can be defined by a normal vector ${}^B\mathbf{n}_p$ and a point ${}^B\mathbf{T}_p$ on the plane in robot base coordinate system. There are three degrees of freedom of a plane in space, and, in general, the plane is also to be defined as a vector

$${}^B\mathbf{a}_p = ({}^B\mathbf{n}_p \cdot {}^B\mathbf{T}_p) {}^B\mathbf{n}_p. \quad (6)$$

This vector also a point on the plane and it is orthogonal to this plane.

III. CALIBRATION METHODS

The objective of calibration is to reduce the systematic errors by correcting parameters introduced in the previous section. In this section, the calibration methods of an eye-to-hand system are described in detail. A close-form solution is then proposed. And finally a solution of optimization is presented.

A. Calibration with a Laser Pointer and a Plane

The overall configuration of an eye-to-hand system is shown in Fig.1 in which a laser pointer is attached on the end-effector and a planar object is placed in front of the camera. The robot base coordinate system is defined as world coordinate system in the following.

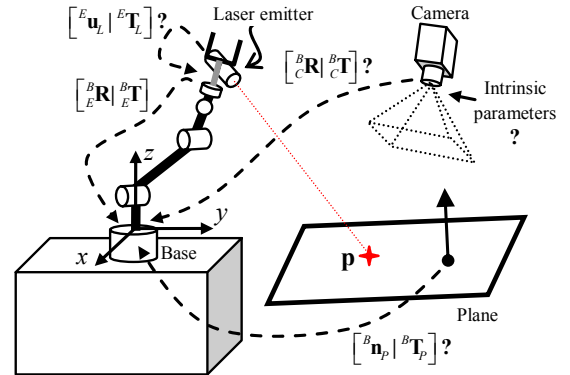


Fig. 1. Overview of an eye-to-hand system with calibration devices.

In the coordinate of the end-effector, the direction of the laser beam is described as a unit vector ${}^E\mathbf{u}_L$. It has only two degrees of freedom and is denoted by Euler angles as

$${}^E\mathbf{u}_L = [\cos {}^E\theta_L \sin {}^E\phi_L \quad \sin {}^E\theta_L \sin {}^E\phi_L \quad \cos {}^E\phi_L]^T \quad (7)$$

where ${}^E\theta_L \in [0, 2\pi)$ is z-axis rotation angle and ${}^E\phi_L \in [0, \pi]$ is angle between z-axis and ${}^E\mathbf{u}_L$. A target point is created when the beam intercepts with a plane. The target point is denoted as ${}^E\mathbf{T}_L$. Since ${}^E\mathbf{T}_L$ could be any point on the line, it only has two degree of freedom. Without loss of generality, let ${}^E\mathbf{T}_L$ be

the point that laser beam intersects with the x - y plane of end-effector coordinate system, i.e.,

$${}^E\mathbf{T}_L \triangleq [0 \ y \ z]_L^T. \quad (8)$$

In the following context, the laser beam is assumed not parallel to the z -axis of the robot. Otherwise, let ${}^E\mathbf{T}_L \triangleq [x \ 0 \ z]_L^T$ or ${}^E\mathbf{T}_L \triangleq [x \ y \ 0]_L^T$. These values are extra parameters which need to be calibrated. Since the laser pointer is rigidly installed, these parameters do not change under normal operation. The 3D position of the spot on the plane is derived by the principle of intersection of a line and a plane in 3D space. The geometric constraint is

$$(d_L {}^B\mathbf{u}_L + {}^B\mathbf{T}_L - {}^B\mathbf{a}_P) \cdot {}^B\mathbf{a}_P = 0 \quad (9)$$

where d_L is the distance from the origin of the laser to the projected point and

$$d_L = \frac{\|{}^B\mathbf{a}_P\| - {}^B\mathbf{T}_L^T \cdot {}^B\mathbf{a}_P}{{}^B\mathbf{u}_L^T \cdot {}^B\mathbf{a}_P}. \quad (10)$$

Hence, the 3D position of the laser spot is described as,

$${}^B\mathbf{p}_{spot} = d_L {}^B\mathbf{u}_L + {}^B\mathbf{T}_L. \quad (11)$$

B. Close-form solution

A close-form solution is derived in this section by exploring special arrangements of the setup. Since no prior guess of parameters is necessary, the proposed method can be run automatically.

Manipulating the end-effector without rotation can generate the constraint equations coming from parallel laser beams. As shown in Fig. 3, under m -th orientation, the end-effector with the laser is moved in a planar space composed of two axes, $\mathbf{i}_m^x$ and $\mathbf{i}_m^y$, and the laser beam is projected on a plane. Rotating the plane and repeating the robot command can make the same laser beams projecting on different planes in space. The working plane can be any one of these planes.

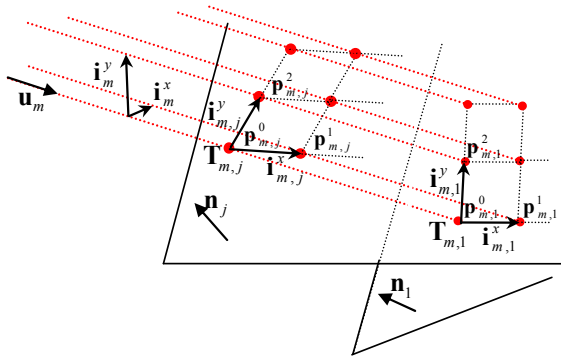


Fig. 3. Projection of parallel laser beams on planes

The effect of lens distortion is ignored in the close-form solution. In other words, the pin-hole camera model of (1) with (3) is adopted. A laser point (X, Y) related with its image position is

$$w\tilde{\mathbf{m}}_{m,j} = \mathbf{H}_{m,j}\tilde{\mathbf{M}} \quad (12)$$

where $\tilde{\mathbf{M}} = [X \ Y \ 1]^T$ and the homography matrix is

defined as

$$\mathbf{H}_{m,j} = \mathbf{K} \begin{bmatrix} {}^C\mathbf{i}_{m,j}^x & {}^C\mathbf{i}_{m,j}^y & {}^C\mathbf{T}_{m,j} \end{bmatrix} \quad (13)$$

This close-form solution is derived from the aspect of measurements of camera to the aspect of movements of the robot.

1) *Find homography matrixes each with an unknown scale factor*: The elements of a homography matrix can be solved by eliminating w according to direct linear transformation (DLT) method [12][13]. Let $\bar{\mathbf{h}}^a$ be the a -th row of $\mathbf{H}$, equation (12) can be rewritten as

$$\mathbf{Q}_H \cdot \mathbf{x}_H = \begin{bmatrix} \tilde{\mathbf{M}}_k^T & \mathbf{0}^T & -u_k \tilde{\mathbf{M}}_k^T \\ \mathbf{0}^T & \tilde{\mathbf{M}}_k^T & -v_k \tilde{\mathbf{M}}_k^T \end{bmatrix}_{m,j} \cdot \begin{bmatrix} \bar{\mathbf{h}}^1{}^T \\ \bar{\mathbf{h}}^2{}^T \\ \bar{\mathbf{h}}^3{}^T \end{bmatrix}_{m,j} = \mathbf{0} \quad (14)$$

By given K points in (m,j) -th frame, $\mathbf{Q}_H$ becomes a $2K \times 9$ matrix. The solution $\mathbf{x}_H$ lies in the null space of the matrix $\mathbf{Q}_H$ and can be solved by calculating the eigenvector corresponding to the smallest eigenvalue of matrix $\mathbf{Q}_H^T \mathbf{Q}_H$. The estimated matrix $\hat{\mathbf{H}}_{m,j}$ with an unknown scaling factor is equal to the homography:

$$\mathbf{H}_{m,j} = \lambda_{m,j} \hat{\mathbf{H}}_{m,j} \quad (15)$$

These scaling factors $\lambda_{m,j}$ will be obtained in the following steps.

2) *Find ratios of the scales of homography matrixes*: Since the laser is moved without rotation in one orientation, the laser beams, $\mathbf{u}_{m,k}$ ($k=1, \dots, K$), are parallel to each other. The vector from $\mathbf{p}_{m,j}^0$ to $\mathbf{p}_{m,1}^0$ is parallel to the vector from $\mathbf{p}_{m,j}^1$ to $\mathbf{p}_{m,1}^1$ and is also parallel to the vector from $\mathbf{p}_{m,j}^2$ to $\mathbf{p}_{m,1}^2$. This means that

$$\left(\begin{bmatrix} \mathbf{K}^{-1}\mathbf{H}_{m,j} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \end{bmatrix} - \begin{bmatrix} \mathbf{K}^{-1}\mathbf{H}_{m,k} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \end{bmatrix} \right) // \left(\begin{bmatrix} \mathbf{K}^{-1}\mathbf{H}_{m,j} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \end{bmatrix} - \begin{bmatrix} \mathbf{K}^{-1}\mathbf{H}_{m,k} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \end{bmatrix} \right) \quad (16)$$

Let $s_{m,j}$ be the ratio between $\lambda_{m,j}$ and $\lambda_{m,1}$:

$$s_{m,j} = \lambda_{m,j} / \lambda_{m,1}. \quad (17)$$

From (16), a constraint equation is

$$(\hat{\mathbf{h}}_{m,1}^1 - s_{m,j} \hat{\mathbf{h}}_{m,j}^1) \times (\hat{\mathbf{h}}_{m,1}^2 - s_{m,j} \hat{\mathbf{h}}_{m,j}^2) = \mathbf{0} \quad (18)$$

and, in the similar way, another constraint is

$$(\hat{\mathbf{h}}_{m,1}^2 - s_{m,j} \hat{\mathbf{h}}_{m,j}^2) \times (\hat{\mathbf{h}}_{m,1}^3 - s_{m,j} \hat{\mathbf{h}}_{m,j}^3) = \mathbf{0} \quad (19)$$

Equations (18) and (19) can be rewritten as,

$$\begin{aligned} \mathbf{A}_{m,j} \mathbf{s}_{m,j} &\triangleq \begin{bmatrix} \hat{\mathbf{h}}_{m,1}^1 \times \hat{\mathbf{h}}_{m,j}^2 + \hat{\mathbf{h}}_{m,1}^2 \times \hat{\mathbf{h}}_{m,j}^1 & -\hat{\mathbf{h}}_{m,1}^1 \times \hat{\mathbf{h}}_{m,j}^3 \\ \hat{\mathbf{h}}_{m,1}^1 \times \hat{\mathbf{h}}_{m,j}^2 + \hat{\mathbf{h}}_{m,1}^2 \times \hat{\mathbf{h}}_{m,j}^1 & -\hat{\mathbf{h}}_{m,1}^2 \times \hat{\mathbf{h}}_{m,j}^3 \end{bmatrix} \begin{bmatrix} s^1 \\ s^2 \end{bmatrix}_{m,j} \\ &= \mathbf{b}_{m,j} \triangleq \begin{bmatrix} \hat{\mathbf{h}}_{m,1}^1 \times \hat{\mathbf{h}}_{m,1}^3 \\ \hat{\mathbf{h}}_{m,1}^2 \times \hat{\mathbf{h}}_{m,1}^3 \end{bmatrix} \end{aligned} \quad (20)$$

Equation (20) can be solved by the least squares method and

$s_{m,j} = s_{m,j}^1$. From this result, the laser beam under the m -th orientation projects on image with a scale is

$$\tilde{\mathbf{u}}_m = \mathbf{K} \cdot (\mu_m^C \mathbf{u}_m) = \hat{\mathbf{h}}_{m,1}^3 - s_{m,j} \hat{\mathbf{h}}_{m,j}^3 \quad (21)$$

where ${}^C \mathbf{u}_m$ is laser beam direction in camera frame.

3) *Find the intrinsic matrix and the hand-to-eye rotation matrix:* The translating axis ${}^B \mathbf{i}_m^x$, the laser beam ${}^B \mathbf{u}_m$ and the projected axis ${}^B \mathbf{i}_m^x$ are lying on the same plane in space, so are ${}^B \mathbf{i}_m^y$, ${}^B \mathbf{u}_m$ and ${}^B \mathbf{i}_m^y$. These imply

$${}^B \mathbf{i}_m^x \cdot ({}^B \mathbf{R} \mathbf{K}^{-1} \tilde{\mathbf{u}}_m \times {}^B \mathbf{R} \mathbf{K}^{-1} \hat{\mathbf{h}}_{m,1}^1) = 0 \quad (22)$$

and

$${}^B \mathbf{i}_m^y \cdot ({}^B \mathbf{R} \mathbf{K}^{-1} \tilde{\mathbf{u}}_m \times {}^B \mathbf{R} \mathbf{K}^{-1} \hat{\mathbf{h}}_{m,1}^2) = 0 \quad (23)$$

These equations can be derived into another from:

$$\mathbf{i}^T \cdot (\mathbf{G}^{-1} \tilde{\mathbf{u}} \times \mathbf{G}^{-1} \hat{\mathbf{h}}) = \begin{matrix} xy & xz & yz \\ xy & xz & yz \end{matrix} \mathbf{w} \cdot \mathbf{g}_{31} + \begin{matrix} xz & yz \\ xz & yz \end{matrix} \mathbf{w} \cdot \mathbf{g}_{21} + \begin{matrix} xy & yz \\ xy & yz \end{matrix} \mathbf{w} \cdot \mathbf{g}_{11} + \begin{matrix} xy & xz & yz \\ xy & xz & yz \end{matrix} \mathbf{w} \cdot \mathbf{g}_{32} + \begin{matrix} xz & yz \\ xz & yz \end{matrix} \mathbf{w} \cdot \mathbf{g}_{22} \quad (24)$$

$$+ \begin{matrix} yz & yz \\ yz & yz \end{matrix} \mathbf{w} \cdot \mathbf{g}_{12} + \begin{matrix} xy & xz & yz \\ xy & xz & yz \end{matrix} \mathbf{w} \cdot \mathbf{g}_{33} + \begin{matrix} xz & yz \\ xz & yz \end{matrix} \mathbf{w} \cdot \mathbf{g}_{23} + \begin{matrix} yz & yz \\ yz & yz \end{matrix} \mathbf{w} \cdot \mathbf{g}_{13} = 0$$

in which some subscripts are ignored and the notification ${}^{abc}_{def} \mathbf{w}$ is defined as ${}^{abc}_{def} \mathbf{w} = (i^a \tilde{u}^b \hat{h}^c - i^d \tilde{u}^e \hat{h}^f)$. For M orientations, we have

$$\mathbf{Q}_G \mathbf{x}_G \triangleq \begin{bmatrix} \begin{matrix} xy & xz & yz \\ xy & xz & yz \end{matrix} \mathbf{w}_1^{x,1} & \dots & \begin{matrix} yz & yz \\ yz & yz \end{matrix} \mathbf{w}_1^{x,1} \\ \begin{matrix} xy & xz & yz \\ xy & xz & yz \end{matrix} \mathbf{w}_1^{y,2} & \dots & \begin{matrix} yz & yz \\ yz & yz \end{matrix} \mathbf{w}_1^{y,2} \\ \vdots & & \vdots \\ \begin{matrix} xy & xz & yz \\ xy & xz & yz \end{matrix} \mathbf{w}_M^{x,1} & \dots & \begin{matrix} yz & yz \\ yz & yz \end{matrix} \mathbf{w}_M^{x,1} \\ \begin{matrix} xy & xz & yz \\ xy & xz & yz \end{matrix} \mathbf{w}_M^{y,2} & \dots & \begin{matrix} yz & yz \\ yz & yz \end{matrix} \mathbf{w}_M^{y,2} \end{bmatrix} \mathbf{x}_G = \mathbf{0}. \quad (25)$$

The solution $\mathbf{x}_G$ lies in the null space of the matrix $\mathbf{Q}_G$. The solution can be solved by calculating the eigenvector corresponding to the smallest eigenvalue of matrix $\mathbf{Q}_G^T \mathbf{Q}_G$. There are 8 degrees of freedom in $\mathbf{x}_G$, so a minimal 4 sets of data to form 8 equations in (25) are needed for the solution. Since each row of the rotation matrix ${}^C_B \mathbf{R}$ is a unit vector, and the last element of $\mathbf{K}$ is 1, the solution is scaled to satisfy $\mathbf{g}_{31}^2 + \mathbf{g}_{32}^2 + \mathbf{g}_{33}^2 = 1$. By applying QR-decomposition, the homogeneous matrix $\mathbf{G}$ is separated into the intrinsic matrix $\mathbf{K}$ and the rotation matrix ${}^C_B \mathbf{R}$.

4) *Find direction of the laser with respect to the end-effector:* From (21),

$$\tilde{\mathbf{u}}_m = \mu_m \mathbf{K}^C {}^B \mathbf{R} {}^B \mathbf{R}_m^E \mathbf{u}_L \quad (26)$$

, i.e.,

$${}^B \mathbf{R}_m^E \mathbf{u}_L = (\mathbf{K}^C {}^B \mathbf{R})^{-1} \tilde{\mathbf{u}}_m / \left\| (\mathbf{K}^C {}^B \mathbf{R})^{-1} \tilde{\mathbf{u}}_m \right\| \quad (27)$$

For M orientations, the M equations are formed and then the laser direction ${}^E \mathbf{u}_L$ is solved by the least squares method.

5) *Find translations of the laser and the camera and the scaling factors:* Define the following relations:

${}^C \mathbf{n}_m^x \triangleq ({}^C \mathbf{i}_m^x \times {}^C \mathbf{u}_m)$ and ${}^C \mathbf{n}_m^y \triangleq ({}^C \mathbf{i}_m^y \times {}^C \mathbf{u}_m)$. Then we have

$${}^C \mathbf{n}_m^{xT} ({}^C \mathbf{T}_{L,m,1} - \lambda_{m,1} \hat{\mathbf{T}}_{m,1}) = 0 \text{ and } {}^C \mathbf{n}_m^{yT} ({}^C \mathbf{T}_{L,m,1} - \lambda_{m,1} \hat{\mathbf{T}}_{m,1}) = 0$$

where $\hat{\mathbf{T}}_{m,1} = \mathbf{K}^{-1} \hat{\mathbf{h}}_{m,1}^3$. Expand these equations into,

$${}^C \mathbf{n}_m^{xT} ({}^C_B \mathbf{R} ({}^B {}^E \mathbf{T}_L + {}^B {}^E \mathbf{T}_{m,1}) + {}^C_B \mathbf{T} - \lambda_{m,1} \hat{\mathbf{T}}_{m,1}) = 0 \quad (28)$$

and

$${}^C \mathbf{n}_m^{yT} ({}^C_B \mathbf{R} ({}^B {}^E \mathbf{T}_L + {}^B {}^E \mathbf{T}_{m,1}) + {}^C_B \mathbf{T} - \lambda_{m,1} \hat{\mathbf{T}}_{m,1}) = 0 \quad (29)$$

Let

$${}^E \mathbf{n}_m^{xT} \triangleq \begin{bmatrix} {}^E n_m^{1,x} & {}^E n_m^{2,x} & {}^E n_m^{3,x} \end{bmatrix} = {}^C \mathbf{n}_m^{xT} {}^C_B \mathbf{R} {}^B \mathbf{R}_m^E \quad (30)$$

and

$${}^E \mathbf{n}_m^{yT} \triangleq \begin{bmatrix} {}^E n_m^{1,y} & {}^E n_m^{2,y} & {}^E n_m^{3,y} \end{bmatrix} = {}^C \mathbf{n}_m^{yT} {}^C_B \mathbf{R} {}^B \mathbf{R}_m^E \quad (31)$$

According to (8), equation (28) and (29) are rewritten as

$$\begin{bmatrix} {}^E n_m^{2,x} & {}^E n_m^{3,x} & {}^C \mathbf{n}_m^{xT} & -{}^C \mathbf{n}_m^{xT} \hat{\mathbf{T}}_{m,1} \\ {}^E n_m^{2,y} & {}^E n_m^{3,y} & {}^C \mathbf{n}_m^{yT} & -{}^C \mathbf{n}_m^{yT} \hat{\mathbf{T}}_{m,1} \end{bmatrix} \begin{bmatrix} {}^E y_L \\ {}^E z_L \\ {}^C_B \mathbf{T} \\ \lambda_{m,1} \end{bmatrix} = \begin{bmatrix} -{}^C \mathbf{n}_m^{xT} {}^C_B \mathbf{R} {}^B {}^E \mathbf{T}_{m,1} \\ -{}^C \mathbf{n}_m^{yT} {}^C_B \mathbf{R} {}^B {}^E \mathbf{T}_{m,1} \end{bmatrix} \quad (32)$$

and the M sets of data form a simultaneous linear equation of $2M$ equations. This can be solved by the least square method, and then the position of the laser in the end-effector frame and the position of the camera in the robot base frame are obtained. The homography matrixes related to the robot dimension are calculated as $\mathbf{H}_{m,1} = \lambda_{m,1} \hat{\mathbf{H}}_{m,1}$ and $\mathbf{H}_{m,j} = \lambda_{m,1} s_{m,j} \hat{\mathbf{H}}_{m,j}$.

6) *Find planes:* The axes and the points on the planes can be obtained from homography matrixes at any orientation and j -th plane as ${}^C \mathbf{i}_{m,j}^x = \mathbf{K}^{-1} \mathbf{h}_{m,j}^1$, ${}^C \mathbf{i}_{m,j}^y = \mathbf{K}^{-1} \mathbf{h}_{m,j}^2$, ${}^C \mathbf{T}_{m,j} = \mathbf{K}^{-1} \mathbf{h}_{m,j}^3$,

${}^B \mathbf{n}_j = {}^B \mathbf{R} ({}^C \mathbf{i}_{m,j}^x \times {}^C \mathbf{i}_{m,j}^y)$ and ${}^B \mathbf{T}_j = {}^B \mathbf{R} {}^C \mathbf{T}_{m,j}$. Finally, according to (6), ${}^B \mathbf{a}_j$'s are obtained.

C. Nonlinear Optimization

The close-form solution discussed above provides a good estimate. However, the error could propagate among steps of the procedure. A nonlinear optimization problem is introduced in this section to refine the results. The laser projecting point is constrained on the plane under different arm poses and this yields the optimization problem as

$$\min_S \sum_{j=1}^J \sum_{k=1}^K \left\| \mathbf{m}_{j,k} - \hat{\mathbf{m}}(S, {}^B \mathbf{R}_{j,k}, {}^B \mathbf{T}_{j,k}) \right\|^2 \quad (33)$$

where ${}^B \mathbf{R}_{j,k}$ and ${}^B \mathbf{T}_{j,k}$ is obtained from k -th command of the manipulator when the plane is put at j -th pose, $\mathbf{m}_{j,k}$ is the estimated point position from (j,k) -th image, $\hat{\mathbf{m}}(\cdot)$ is the prediction of the projection position in the image according to (1) to (11), and S is the set of parameters to refine, including f_u , f_v , α_c , u_0 , v_0 , κ_1 , κ_2 , ρ_1 , ρ_2 , ${}^C_B \theta$, ${}^C_B \phi$, ${}^C_B \varphi$, ${}^C_B \mathbf{T}$, ${}^E \theta_L$, ${}^E \phi_L$, ${}^E y_L$, ${}^E z_L$, and ${}^B \mathbf{a}_j$, $j = 1, 2, \dots, J$.

Equation (33) is a nonlinear minimization problem which

can be solved with the Levenberg-Marquardt method. It requires an initial guess of S . The initial values can be obtained from the previously described close-form solution. Secondly, if all the parameters of the system can be obtained from specifications or design settings, these values should be close to real ones and can be used as initial guess in the for optimization.

IV. EXPERIMENTAL RESULTS

The proposed algorithm is evaluated by simulations and real data in this section.

A. Simulations

In order to validate the proposed method, simulations are performed with different sample number, under different noise level and different camera position. The real values of parameters in all cases are set as following: $f_u=543$, $f_v=543$, $\alpha_c=0.05$, $u_0=320$, $v_0=240$, ${}^B_c\theta=-90^\circ$, ${}^B_c\phi=-10^\circ$, ${}^B_c\varphi=-170^\circ$, ${}^B_cT=[510, 151, 430]^T$ (mm), ${}^E\theta_L=21.80^\circ$, ${}^E\phi_L=53.39^\circ$, ${}^E\mathbf{T}_L=[0, -10, 30]^T$ (mm), ${}^B\mathbf{a}_1=[12.82, -1.90, -68.71]^T$ (mm), and ${}^B\mathbf{a}_2=[124.47, 15.02, -226.74]^T$ (mm).

We test the effect of noise in the point positions in the images. For the motion of the manipulator, seven random orientations each with $25(5 \times 5)$ positions in random axes are generated to project laser on the two planes and then produce a total of 250 samples with measurements of the camera. Every measurement is contaminated by a Gaussian noise with zero mean and a standard deviation. These data are both applied to the close-form solution and nonlinear optimization. The estimated parameters are compared to the real ones and their relative errors in percentage are calculated. In this test, the noise level (i.e. standard deviation) in pixel is varying from 0 to 1 pixel. 100 independent trials are performed for each noise level and the averaged results are shown in Fig.5. Fig.5(a) is the result of the close-form solution and Fig.5(b) is the result refined by the nonlinear optimization method. The average error increase when the noise level increases. The curves of relative errors are almost linear in close-form solution. The errors of optimization are less than 1% when the noise level is less than 0.3 pixel.

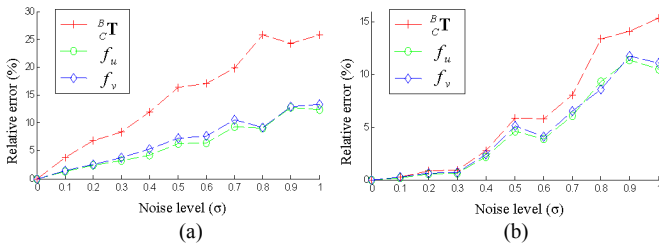


Fig. 5. Relative errors with respect to the noise level. (a) Closed-form solution and (b) Nonlinear optimization.

Further, we test the performance with different numbers of samples. Zero mean Gaussian noise with 0.5 pixel standard deviation is added in the point positions in the images. For the motion of the manipulator, 5 to 14 random orientations each

with $25(5 \times 5)$ positions in random axes are generated to project laser on the 2 planes and then produce a total of 250 to 700 samples with measurements of camera projections. For each number, 100 independent trials are performed and the averages of errors are shown in Fig.5. The result shows that the errors decrease as the number increases. The nonlinear optimization provides a better result that has about 50% less errors than the close-form solution does.

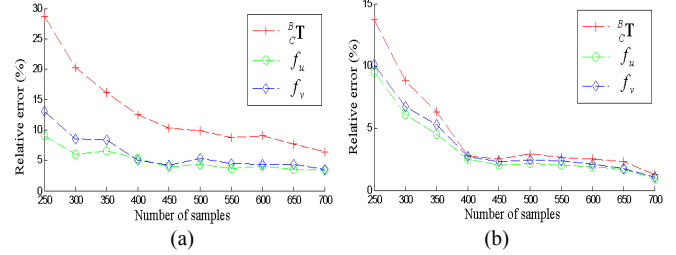


Fig. 6. Relative errors with respect to the number of samples. (a) Closed-form solution and (b) Nonlinear optimization.

B. Real data

The experimental setup is shown in Fig.7. The manipulator is TX60 with a controller, CS8, manufactured by Stäubli Inc. The camera is Panasonic WV-CP 480 CCD camera with a 8mm lens. A planar plate of size 50×50 cm is placed about 90cm away from the manipulator. In this configuration, the camera can not see any part of the manipulator. The laser pointer is a laser diode with an adjustable lens.

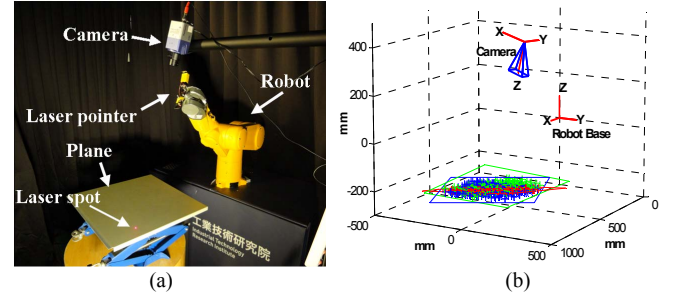


Fig. 7. (a) Experimental setup. (b) Extrinsic results.

We manipulated the end-effector to $25(5 \times 5)$ positions in a planar space composed of two different directions under the same orientation and repeated Step 3 to 4 four times for different orientations and recorded every pose. We collected 500 samples in Step 5 for optimization. All movements were designed to project the laser beam on the planar object. These poses were repeated with total three different orientations. The data from Step 3 were applied to the proposed close-form solution to obtain initial values and then nonlinear optimization method was utilized to refine values. The results are shown in Table I and there are four tests in this experiment. The first column in the first test, labeled *Initial*, is the estimation of the close-form solution. The second column, labeled *Refined*, is the estimation of the nonlinear optimization and the third column is the standard deviation, indicating the uncertainty of the refined result. Fig. 8 shows the extrinsic results of the first test. Second test apply more

samples, including the 375 samples of initialization, to refine the values from the first test. For the third and the fourth tests, the data are selected by outlier rejection according to the difference of the estimated position and the predicted one using the refined parameters of first test. The threshold is set to be 2.5 pixel. In each test, the last row, labeled by RMS, shows the root mean square of the distance of image from the estimated points to the predicted points. The last column shows the intrinsic parameters estimated by applying *Camera Calibration Toolbox for Matlab* [14]. We performed the camera calibration with a pattern of 35(5×7) grids. The grid size is 15mm×15mm. 50 images of the pattern under different orientations were taken. The values of the last column are close to previous ones. From Table I, we can see that the results are consistent with each other; moreover, the uncertainty of estimations and RMS value declines as the data number increases and are significantly lower in the results with outlier rejection.

V. CONCLUSION

This work develops a calibration algorithm to accurately estimate the camera intrinsic parameters, the configuration between a camera and a robot, and the pose of working plane at the same time. This technique is accomplished through a planar object and a laser pointer. Since the laser is rigidly installed and the planar object is fixed, the measurements of the camera that the laser beam projected on the plane should obey the constraint in geometric relation. Based on the constraint, the proposed close-form solution is to obtain initial values and the nonlinear optimization can refine the values. Computer simulations evaluate the calibration method and analyze the performances in different conditions. The experiment proofs that this method is useful when the camera can not see the hand and result with real data is consistent with the simulation and validates the proposed algorithm.

REFERENCES

- [1] G. Flandin, F. Chaumette, and E. Marchand, "Eye-in-hand/eye-to-hand cooperation for visual servoing," in *Proc. IEEE Int. Conf. Robotics and Automation*, pp. 2741-2746, San Francisco, April 2000.
- [2] S. Hutchinson, G. D. Hager, and P. I. Corke, "A tutorial on visual servo control," *IEEE Trans. Robotics and Automation*, vol. 12, pp. 651-670, Oct. 1996.
- [3] Y. C. Shiu and S. Ahmad, "Calibration of Wrist-Mounted Robotic Sensors by Solving Homogeneous Transform Equations of the Form $AX = XB$," *IEEE Trans. Robotics and Automation*, vol. 5, no. 1, pp. 16-29, Feb. 1989.
- [4] R. Y. Tsai and R. K. Lenz, "A New Technique for Fully Autonomous and Efficient 3D Robotics Hand/eye Calibration," *IEEE Trans. Robotics and Automation*, vol. 5, no. 3, pp. 345-358, June 1989.
- [5] F. Dornaika and R. Horaud, "Simultaneous Robot-World and Hand-Eye Calibration," *IEEE Trans. Robotics and Automation*, vol. 14, no. 4, pp. 617-622, Aug. 1998.
- [6] N. Andreff, R. Horaud, and B. Espiau, "Robot Hand-Eye Calibration Using Structure-from-Motion," *Int. J. Robotics. Research*, vol. 20, no. 3, pp. 228-248, March 2001.
- [7] K. H. Strobl and G. Hirzinger, "Optimal Hand-Eye Calibration," in *Proc. of the IEEE/RSJ Int. Conf. Intelligent Robots and Systems*, pp. 4647-4653, Beijing, China, Oct. 2006.
- [8] H. Xu, Y. Wang, W. Chen and J. Lu, "A Self-Calibration Approach to Hand-Eye Relation Using a Single Point," in *Proc. of the IEEE Int. Conf. Information and Automation*, pp. 413-418, Zhangjiajie, China, June 2008.
- [9] K. H. Strobl and G. Hirzinger, "More Accurate Camera and Hand-Eye Calibrations with Unknown Grid Pattern Dimensions," in *Proc. IEEE Int. Conf. Robotics and Automation*, Pasadena, CA, USA, pp. 1398-1405, May 2008.
- [10] S. Lei, L. Jingtai, S. Weiwei, W. Shuihua, and H. Xingbo, "Geometry based robot calibration method," in *Proc. IEEE Int. Conf. Robotics and Automation*, pp. 1907-1912, New Orleans, LA, April 2004.
- [11] J.S. Hu, M.C. Chien, Y.J. Chang, Y.C. Chang, S.H. Su, J.J. Yang, and C.Y. Kai, "A Robotic Ball Catcher with Embedded Visual Servo Processor," in *Proc. of the IEEE/RSJ Int. Conf. Intelligent Robots and Systems*, to be published.
- [12] R. I. Hartley and A. Zisserman, *Multiple View Geometry in computer vision*, Cambridge University Press, 2003.
- [13] Z. Zhang, "A Flexible New Technique for Camera Calibration," *IEEE Trans. Pattern Analysis and Machine Intelligence*, vol. 22, no. 11, pp. 1330-1334, Nov. 2000.
- [14] J.-Y. Bouguet. Camera calibration toolbox for Matlab. http://www.vision.caltech.edu/bouguetj/calib_doc/.

TABLE I
RESULTS WITH REAL DATA OF DIFFERENT NUMBER OF SAMPLES

Parameter	375 samples (125 samples per plane)			525 samples (175 samples per plane)		Unit
	Initial	Refined	σ	Refined	σ	
(f_u, f_v, α_c)	(835.73, 863.81, 0.06)	(1055.85, 1023.69, -0.01)	(39.61, 40.59, 0.01)	(1047.72, 1029.89, 0.00)	(24.58, 24.79, 0.00)	pixel
(u_0, v_0)	(137.22, 142.13)	(374.03, 262.59)	(33.56, 25.05)	(342.86, 246.26)	(19.39, 15.36)	pixel
(k_1, k_2)	(0, 0)	(-0.3170, -0.2251)	(0.0721, 0.4677)	(-0.3176, -0.2267)	(0.0616, 0.4506)	1
(p_1, p_2)	(0, 0)	(-0.0040, 0.0032)	(0.0078, 0.0092)	(0.0042, 0.0056)	(0.0042, 0.0045)	1
$({}^B_c\theta, {}^B_c\phi, {}^B_c\varphi)$	(-89.25, -7.94, -174.66)	(-92.95, -2.05, -164.70)	(0.47, 0.81, 1.40)	(-92.77, -3.25, -166.36)	(0.23, 0.56, 0.90)	deg.
${}^B_c\mathbf{T}^T$	[921.74, 165.04, 155.23]	[817.71, 251.36, 461.24]	[13.00, 4.95, 25.88]	[816.32, 250.27, 455.28]	[7.16, 3.93, 14.70]	mm
RMS	174.7426	2.3428		2.1200		pixel
Parameter	525 samples (175 samples per plane)		675 samples (225 samples per plane)		Camera Calibration Toolbox for Matlab	Unit
	Refined	σ	Refined	σ		
(f_u, f_v, α_c)	(1077.76, 1060.82, 0.00)	(13.16, 12.92, 0.00)	(1039.56, 1028.84, 0.00)	(9.34, 9.08, 0.00)	(1039.96, 1040.46, 0)	pixel
(u_0, v_0)	(370.66, 259.68)	(10.03, 7.98)	(321.58, 250.02)	(7.14, 6.04)	(311.66, 211.31)	pixel
(k_1, k_2)	(-0.3112, -0.4220)	(0.0357, 0.2931)	(-0.3895, 0.4891)	(0.0298, 0.2318)	(-0.35336, 0.31727)	1
(p_1, p_2)	(-0.0012, -0.0022)	(0.0022, 0.0023)	(-0.0026, 0.0016)	(0.0016, 0.0014)	(0.0032, 0.0023)	1
$({}^B_c\theta, {}^B_c\phi, {}^B_c\varphi)$	(-92.94, -2.63, -164.99)	(0.13, 0.31, 0.50)	(-92.96, -3.89, -167.88)	(0.09, 0.24, 0.38)	N/A	deg.
${}^B_c\mathbf{T}^T$	[813.03, 253.92, 474.24]	[3.45, 2.76, 7.78]	[808.05, 245.71, 451.60]	[2.66, 1.98, 5.23]	N/A	mm
RMS	1.0294		0.9852			pixel